THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 3

Chapter 13: **Earth as a System: Energy, Matter and Life** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The main source of energy for the Earth system is: 1

(a) the wind **(b)** the Moon **(c)** the Sun **(d)** the core

2. The natural greenhouse effect is: 1

(a) absent **(b)** harmful always

(c) caused only by humans **(d)** necessary for life on Earth

3. UV radiation is harmful because it can cause: 1

(a) good eyesight **(b)** better growth

(c) stronger bones **(d)** skin cancer and cataract

4. The main reservoir of carbon on Earth in living things is: 1

(a) plants and animals **(b)** oxygen

(c) nitrogen **(d)** helium

5. The speed of light in vacuum is about: 1

(a) 3 x 10^3 m/s **(b)** 3 x 10^8 m/s **(c)** 3 x 10^5 m/s **(d)** 340 m/s

6. Which of the following is a greenhouse gas? 1

(a) oxygen **(b)** nitrogen **(c)** carbon dioxide **(d)** argon

7. UV radiation is useful because it: 1

(a) helps make vitamin D and kills germs **(b)** makes ozone

(c) heats the Earth **(d)** causes cancer

8. Carbon dioxide is removed from the air by: 1

(a) photosynthesis **(b)** decay **(c)** burning fuels **(d)** respiration

9. Light from the Sun takes about ___ to reach the Earth. 1

(a) 8 hours **(b)** 8 days **(c)** 8 minutes **(d)** 8 seconds

10. The increase of Earth's average temperature due to excess greenhouse gases is called: 1

(a) ozone depletion **(b)** eutrophication **(c)** global warming **(d)** acid rain

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. What is meant by the greenhouse effect? 2

12. What is the ozone layer? Why is it important? 2

13. Why is India suitable for solar energy? 2

14. Name the main steps of the water cycle. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Describe the nitrogen cycle in brief. 3

16. What is Mission LiFE? State three practices under it. 3

17. Explain why the clouds and the atmosphere affect the solar radiation reaching the ground. 3

OR

Describe the water cycle with the role of the Sun. 3

18. Why is the Sun called the ultimate source of energy for the Earth system? Give three reasons. 3

OR

State three sustainable practices to protect the Earth system. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe the water cycle and explain its role in the Earth system with the processes of evaporation, transpiration, condensation, precipitation, run-off and infiltration. 5

OR

Explain how the atmosphere, hydrosphere, geosphere and biosphere are linked in the processes of weathering, photosynthesis, and the formation of rain. 5

20. Describe the carbon cycle with the processes and the reservoirs and the human impact. 5

OR

Discuss the Earth system's response to human activities such as burning fuels, deforestation, and the use of CFCs with the evidence of global change. 5

END OF QUESTION PAPER — **Total: 40 Marks**

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